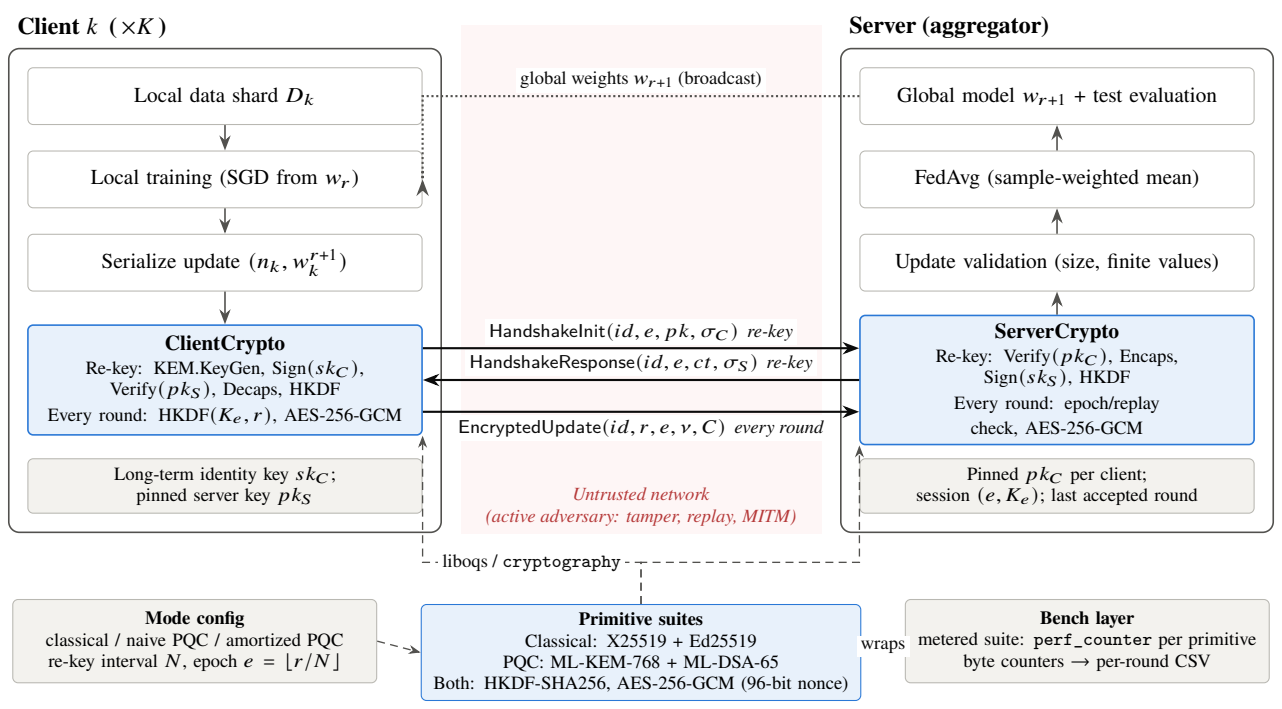
Client k ($\times K$)
Local data shard D_k
Local training (SGD from w_r)
Serialize update (n_k, w_k^{r+1})
ClientCrypto
Re-key: KEM.KeyGen, Sign(sk_C), Verify(pk_S), Decaps, HKDF
Every round: HKDF(K_e, r), AES-256-GCM
Long-term identity key sk_C; pinned server key pk_S
global weights w_{r+1} (broadcast)
HandshakeInit(id, e, pk, σ_C) re-key
HandshakeResponse(id, e, ct, σ_S) re-key
EncryptedUpdate(id, r, e, v, C) every round
Untrusted network
(active adversary: tamper, replay, MITM)
Server (aggregator)
Global model w_{r+1} + test evaluation
FedAvg (sample-weighted mean)
Update validation (size, finite values)
ServerCrypto
Re-key: Verify(pk_C), Encaps, Sign(sk_S), HKDF
Every round: epoch/replay check, AES-256-GCM
Pinned pk_C per client; session (e, K_e); last accepted round
liboqs / cryptography
Mode config
classical / naive PQC / amortized PQC
re-key interval N, epoch $e = \lfloor r/N \rfloor$
Primitive suites
Classical: X25519 + Ed25519
PQC: ML-KEM-768 + ML-DSA-65
Both: HKDF-SHA256, AES-256-GCM (96-bit nonce)
wraps
Bench layer
metered suite: perf_counter per primitive
byte counters → per-round CSV